

Optimizing the Martini 3 Force Field Reveals the Effects of the Intricate Balance between Protein–Water Interaction Strength and Salt Concentration on Biomolecular Condensate Formation

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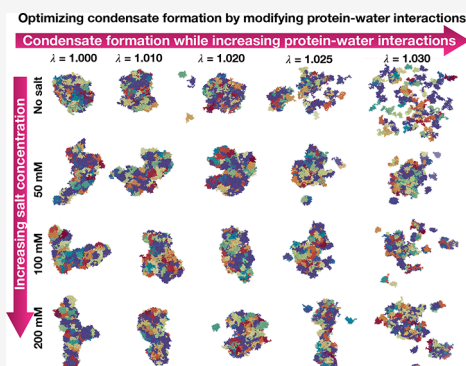


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ABSTRACT: Condensation/dissolution has become a widely acknowledged biological macromolecular assembly phenomenon in subcellular compartmentalization. The MARTINI force field offers a coarse-grained protein model with a resolution that preserves molecular details with an explicit (CG) solvent. Despite its relatively higher resolution, it can still achieve condensate formation in a reasonable computing time with explicit solvent and ionic species. Therefore, it is highly desirable to tune this force field to be able to reproduce the experimentally observed properties of the condensate formation. In this work, we studied the condensate formation of the low-sequence complexity domain of fused in sarcoma protein using a MARTINI 3 force field by systematically modifying (increasing) the protein–water interaction strength and varying the salt concentration. We found that the condensate formation is sensitive both to the protein–water interaction strength and the presence of salt. While the unmodified MARTINI force field yields a complete collapse of proteins into one dense phase (i.e., no dilute phase), we reported a range of modified protein–water interaction strength that is capable of capturing the experimentally found transfer free energy between dense and dilute phases. We also found that the condensates lose their spherical shape upon the addition of salt, especially when the protein–water interactions are weak. Interchain amino acid contact map analysis showed one explanation for this observation: the protein–protein contact fraction reduces as salt is added to systems (when the protein–water interactions are weak), consistent with electrostatic screening effects. This reduction might be responsible for the condensates becoming nonspherical upon the addition of salt by reducing the need for minimizing the interfacial area. However, as the protein–water interactions become stronger to the extent that makes the transfer free energy agree well with experimentally observed transfer free energy, we found an increase in the protein–protein contact fraction upon the addition of salt, consistent with the salting-out effects. Therefore, we concluded that there is an intricate balance between screening effects and salting-out effects upon the addition of salt and this balance is highly sensitive to the strength of protein–water interactions.



INTRODUCTION

The condensation/dissolution phenomenon in subcellular compartmentalization has been brought to the spotlight by the seminal work by Brangwynne et al. for cytoplasmic granules,¹ after which the research field of biomolecular organization exploded with many other examples of subcellular compartments that form via a similar phenomenon.^{2–4} One mechanism that explains this phenomenon is liquid-like phase separation (LLPS),⁵ in which the two liquid-like phases are a protein-rich (condensed or dense) phase and a dilute phase. Phase transitions, like LLPS, are ubiquitous in biological macromolecular assembly, and vital for organizing and regulating cellular biochemical processes.^{6–13}

Molecular simulations of biomolecular condensates have been particularly useful in providing means to directly probe the interactions that mediate the condensate formation. The fundamental trade-off in molecular modeling (i.e., the accessible time and length scales versus the model resolution), however,

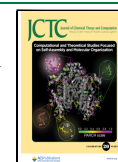
limits the range of applications of a given model, e.g., resolutions lower than atomistic scale are often needed for systematic studies of condensate formation.¹⁴ Accordingly, coarse-grained (CG) models have become a popular option to study condensate formation.^{15–25} Among the CG models, the MARTINI force field offers a resolution that preserves molecular details with an explicit (CG) solvent while being able to achieve condensate formation in a reasonable computing time.^{22,24}

The MARTINI force field was originally developed to reproduce experimental free energies of chemical building

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blocks' partitioning between polar and apolar phases^{26,27} and has been employed successfully for several purposes. Some recent examples include lipid nanoemulsions for drug delivery,²⁸ PEGylation of proteins,²⁹ and predicting the structure of membrane-associated protein assemblies.³⁰ Recently, Benayad et al.²⁴ has shown that an older version of the MARTINI force field (version 2.2) can also successfully reproduce the experimentally observed (*in vitro*) excess transfer energy of the low-sequence complexity domain of FUS protein (FUS LC) between condensed and dilute phases by scaling the protein–protein interactions. The authors obtained further insights into the condensate properties by using this scaled MARTINI 2.2 force field, such as shear viscosity and surface tension of droplets. Moreover, Tsanai et al.²² have shown that an open beta version of MARTINI 3.0 can reproduce the salt-dependent coacervate formation of polyelectrolyte type peptides (by applying elastic bonds between backbone beads to enforce backbone extension). In this work, we aim to optimize an open-beta version of the MARTINI 3.0 force field to reproduce the experimentally found properties of the biomolecular condensate formation.

Single-molecule properties, most notably, the size-related measurables (such as radius of gyration), have been successfully used as objective functions for force field optimization, which yield a reasonable agreement between experiments and simulation data for non-size-related test parameters (such as backbone structure, transport properties, etc.).^{31–34} One of the most commonly used tuning parameters for such force field optimizations is the protein–water interaction strength (specifically, ϵ of Lennard-Jones (LJ) interactions between protein–water atoms/particles). This parameter has been successfully used for the optimization of atomistic force fields^{31,32} as well as the MARTINI force field.^{33,35} Larsen et al.³³ and Thomasen et al.³⁵ have devised this approach for an open-beta version of MARTINI 3.0 (3.0.beta.4.17) and MARTINI 3.0.0 (release version), respectively, where they showed that the overly compact behavior of intrinsically disordered proteins and multidomain proteins can be improved by a 6 to 10% increase in protein–water interactions.

In this work, we have employed a similar approach as in the work by Lindorff-Larsen^{33,35} by systematically modifying the nonbonded (LJ) interaction strength between the protein–water beads, in order to reproduce the condensate properties, specifically the excess free energy of transferring protein chains from dilute phase to condensed phase.²⁴ For this benchmarking study, we used a 163 residue-long peptide (FUS LC) that represents the low sequence-complexity domain of the FUS RNA-binding protein. We presented the results from each scaling at various salt concentrations. We found that the dense phase formation is sensitive to both the strength of the protein–water interaction and the salt concentration. We reported a range of protein–water interaction scaling where we were able to reproduce experimentally observed transfer free energy, although we found that the protein concentration in the dense phase was higher than the experimentally detected concentrations. We also found that the presence of salt concentration affects the dense phase morphology, especially when the protein–protein interactions dominate the protein–water interactions. We showed that the individual amino acid contacts are highly sensitive to salt concentration; salt has more predominant screening effects when protein–water interactions are weak whereas the opposite salting-out effects become more predominant as protein–water interactions strengthen. We conclude that molecular simulations with well-balanced force

fields (i.e., well-balanced protein–protein and protein–water interactions) are highly useful in studying the microscopic details of the interactions governing biomolecular self-assembly.

METHODS

Here we performed coarse-grained molecular dynamics simulations of the FUS LC domain, which is an ideal benchmarking system as it has been shown to form condensates *in vitro*³⁶ and has plenty of experimental and computational data available for comparison.^{36–39} The full sequence of simulated 163 residue-long FUS LC is given in the [Supporting Information \(SI\)](#).

Modeling. We employed an open-beta version of the MARTINI 3.0 force field (3.0.beta.3.2)^{40,41} for modeling the FUS LC domain. No elastic backbone (or side chain) constraints were applied (i.e., FUS LC is kept fully flexible). We kept the protein–protein interactions unmodified but systematically tested a range of scaled protein–water interactions. Specifically, we modified the ϵ of the Lennard-Jones interactions between protein and water beads following a strategy similar to the ones in the single-molecule studies by Lindorff-Larsen and co-workers.^{23,33,35} We will refer to the scaling parameter as λ ; the parameter with which we multiply the ϵ of Lennard-Jones interactions between protein and water beads. We presented the results where the λ parameter varies between 1 and 1.04.

Initial atomic coordinates of FUS LC were prepared as an extended configuration that is energy minimized using Amber99SBdisp force field.⁴² The atomistic coordinates were then converted to coarse-grained coordinates by using the *martinize* script provided within MARTINI 3.0.beta.3.2 distribution.⁴¹ We prepared cubic simulation boxes ($L = 40$ nm) that contained 100 copies of FUS LC chains. This setting gives a protein concentration of ≈ 2.6 mM (44.5 mg/mL). Benayad et al. have shown that this concentration fits within the range that would yield a spherical condensate (instead of a percolated network).²⁴ Moreover, for this volume and concentration, spherical droplet formation simulations and direct coexistence (slab) simulations yield very similar equilibrium densities of the condensed and dilute phases.²⁴

Each of our systems starts from a monodispersed initial condition. A snapshot of an example starting configuration is shown in the SI (Figure S1). The simulation boxes were then solvated in MARTINI water and randomly selected water beads are replaced with ion beads both to provide electroneutrality (a single chain of FUS LC domain has a net negative charge -2) and to adjust the salt concentration (tested 0, 50, 100, and 200 mM NaCl concentrations). We used the relevant TQ bead types for Na^+ and Cl^- ions and kept the ion–water and ion–protein interactions unmodified. Equilibrated initial conditions, topology files, and run parameter files for all λ values of 0 mM and 100 mM salt are publicly available on GitHub (https://github.com/gzerze/martini3.0.beta_FUS_input_files).

Simulation Details. After initial solvation, all systems were energy minimized using the steepest descent algorithm. They were then equilibrated with 400 ps NVT simulations ($T = 300$ K) followed by 400 ps NPT simulations ($T = 300$ K, $P = 1$ bar), where the temperature was maintained constant with the velocity-rescaling algorithm⁴³ a 2 ps time constant and pressure was maintained constant using Parrinello–Rahman barostat⁴⁴ with a time constant of 20 ps. A cutoff distance of 1.1 nm was used both for the van der Waals interactions and short-range electrostatic interactions. Long-range electrostatic interactions

were calculated using a generalized reaction field method.⁴⁵ We note that a long-range treatment of electrostatic interactions is essential to obtain an accurate phase behavior.⁴⁶ The leapfrog algorithm was used to integrate the equations of motion with a time step of 20 fs. All simulations were performed using GROMACS MD engine (version 2016.3).^{47–49}

Cluster Formation Analysis. Cluster formation is defined based on a minimum distance criterion. We first calculated the closest distance between each protein pair as a function of time. Any two protein molecules are considered to be in the same cluster if any two beads of the molecules are within 0.5 nm (or less) distance from each other. Based on this criterion, we built adjacency matrices and then found the connected components by using the compressed sparse graph routines of public Python libraries.⁵⁰ We then labeled the connected components based on their size, the largest one being the largest cluster and so on.

RESULTS AND DISCUSSIONS

Identification of Condensate Formation. We first identified the dense-phase (droplet) formation by analyzing the cluster formation during the simulations. Any two protein molecules are considered to be in the same cluster if any two beads of the molecules are within 0.5 nm (or less) distance from each other (see Methods section for details). We then quantified the fraction of proteins within given clusters. Fractions of proteins in the most populated three clusters are presented in Figure 1. While all the systems start from a randomly monodispersed state, we found that all protein chains coalesce into a single cluster for smaller λ ($\lambda < 1.02$) values after some equilibration time (Figure 1). We determined the equilibration time based on the cluster formation (Figure 1) and the fluctuations in the total potential energy of the system (Figure

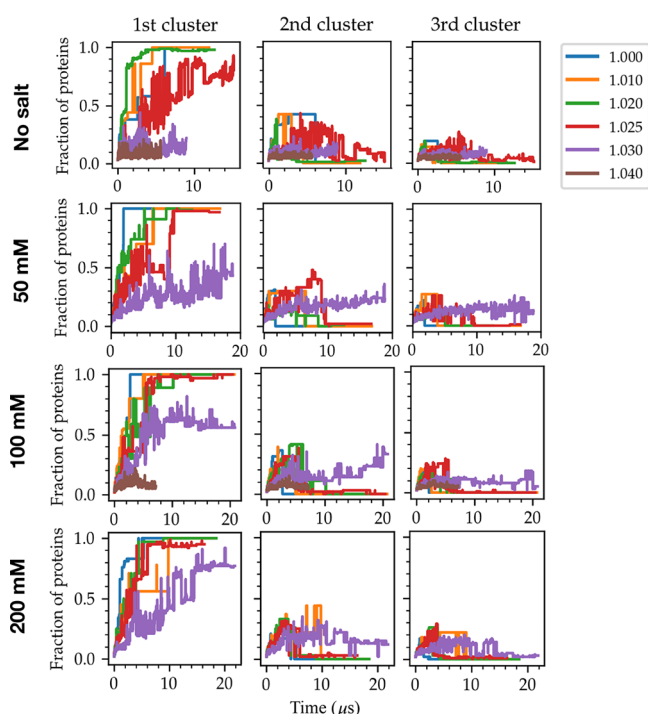


Figure 1. Fraction of proteins in three most populating clusters for the λ values tested and for various salt concentrations (top row: 0 mM salt, second row: 50 mM NaCl, third row 100 mM NaCl, last row: 200 mM NaCl).

S2). Equilibration periods are tabulated in Table S1 for all systems. Representative snapshots of the systems are shown in Figure 2 after the cluster formation was observed (i.e., after the equilibration time).

We labeled the cluster with the largest fraction of proteins as the droplet (or the dense phase) and the remainder of the system as the dilute phase. We did not find any proteins in the dilute phase for $\lambda < 1.02$. At no salt condition, for $\lambda = 1.02$, we found that while the majority of the proteins (97–98%) coalesce into one cluster, a small percentage of proteins (fluctuating between 2 and 3%) is present in the dilute phase (Figure 1). The fraction of proteins in the largest cluster (i.e., in the dense phase) becomes smaller as λ gets larger. At $\lambda = 1.025$ (no salt), this fraction becomes ≈ 0.75 and the droplets start to show significant deviations from a round shape (see the snapshot given in Figure 2, top row, the fourth column from left). Beyond $\lambda = 1.03$, there was not a dominant cluster formation (Figure 1). To further verify that no droplets are stable beyond $\lambda = 1.03$, we also ran $\lambda = 1.04$ simulations at no salt conditions starting from a preformed spherical droplet (last saved configuration of $\lambda = 1$ at 0 mM salt). We then analyzed the cluster formation as a function of time (Figure S3) and found that the initial droplet dissolves quickly (at a time scale shorter than that of the formation of the droplets).

For the conditions where we have a finite salt concentration, we found that clusters are forming beyond the periodic boundaries (percolation) for some cases, especially at small λ (e.g., see Figure 2, second row, the first column from left for $\lambda = 1$ at 50 mM salt concentration), unlike 0 mM salt conditions. These clusters at finite salt concentrations also show substantial deviations from a spherical shape. These observations indicate that the condensate morphologies change significantly upon the addition of salt. We also investigated the time dependence of the size and shape of these morphologies by quantifying the radius of gyration (R_g) and the ratio of the smallest and largest eigenvalues of the gyration tensor (Figure S4). The latter offers a measure of the sphericity of droplets. We found that low λ cases at 0 mM salt ($\lambda = 1, 1.01, 1.02$) have the most spherical morphologies. Beyond $\lambda = 1.025$ at no salt, the cluster formation is not tight (as evident from the R_g) so it also loses its sphericity. The condition that shows percolation ($\lambda = 1$ at 50 mM salt) has the largest deviation from the sphericity (it is rod-like instead) combined with a large R_g .

Recent studies from independent groups show that the nonspherical droplet formation might be a kinetic arrest, playing an important role in droplet maturation and aging.^{51–53} To test whether the nonspherical morphologies that we observed are impacted by the initial conditions, we rerun 50 mM at all λ values starting from a preformed droplet (last saved configuration of 0 mM salt, $\lambda = 1$ condition). We simulated each λ for 10 μ s and presented the analysis in Figure S5. Within the given simulation time, the initially spherical droplets stayed intact and spherical, except for $\lambda = 1.03$, which had one copy of the FUS LC protein exchange back and forth between the dense and dilute phases. The enlarged droplet in the case of $\lambda = 1.03$ also deviated from its initially spherical shape. These findings show that the nonspherical morphology was not reproducible for λ values less than 1.03 when starting from a preformed spherical droplet. We argue that the strength of effective protein–protein interactions at low λ is largely responsible for the initial spherical droplet staying intact. While the further investigation of the changes in the morphology is a subject of future work, here we further analyzed the per-residue contact formation between the protein chains in

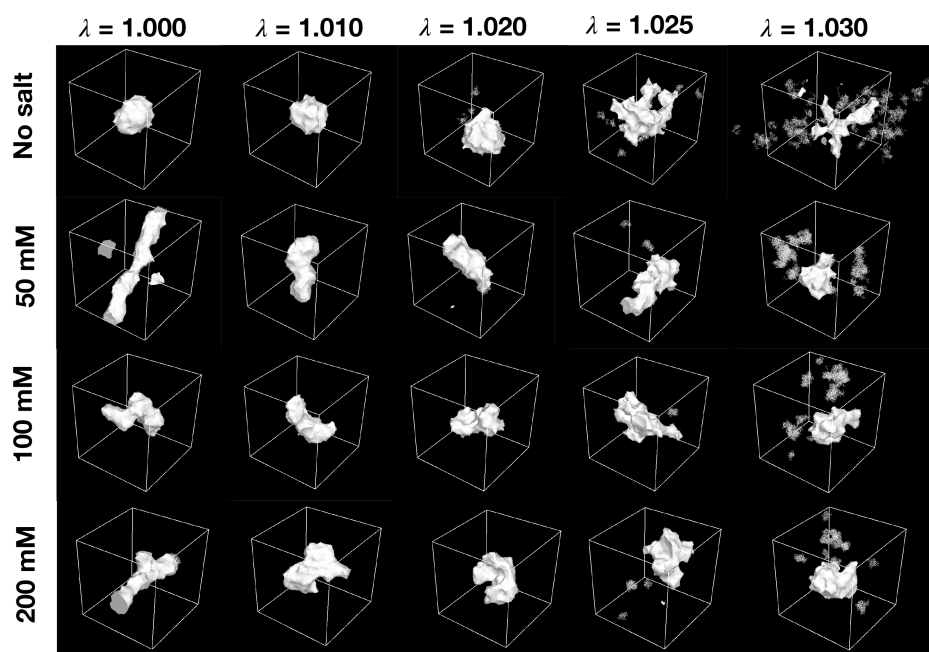


Figure 2. Representative snapshots of the systems for the λ values tested and for various salt concentrations (top row: 0 mM salt, second row: 50 mM NaCl, third row 100 mM NaCl, last row: 200 mM NaCl) obtained after the cluster formation is observed. A surface representation (with a probe radius 8 Å is used to illustrate protein in OVITO. Further details of the surface reconstruction are available in the [Concentration Profiles](#) subsection.

Table 1. Protein Concentrations in the Dense Phase Given in mg/mL^a

		$\lambda = 1.000$	$\lambda = 1.010$	$\lambda = 1.020$	$\lambda = 1.025$	$\lambda = 1.030$
no salt	sigmoidal fit	1328.7	1250.4	964.2	720.8 ± 12.4	—
	surface reconstruction	1311.2	1213.2	1054.4	999.8 ± 18.1	—
50 mM	surface reconstruction	1327.9	1271.2	1164.7	1135.5 ± 17.5	1144.2 ± 15.8
100 mM	surface reconstruction	1310.4	1259.0	1179.5	1141.3 ± 19.4	1110.2 ± 12.0
200 mM	surface reconstruction	1287.9	1235.9	1161.8	1106.4 ± 15.2	1069.4 ± 12.6

^aFor the no-salt condition, the concentrations are obtained from two different techniques (sigmoidal fit and surface reconstruction, see the text for more details). Statistical errors calculated by block averaging of the data (dividing the equilibrated data into two equal blocks) are less than 1% at low λ conditions. Errors larger than 1% are reported.

the [Contact Maps](#) subsection, which could shed some light on the impacted morphologies.

Concentration Profiles. In order to analyze the partitioning of individual species into the condensed and dilute phases, we quantified the average radial concentration profiles of the species. For this analysis, we defined the center as the center of the condensate; calculated the concentration profiles for each instantaneous frame (after equilibration); and calculated the ensemble average. While the protein concentration profile for no salt conditions follows a sigmoidal trend, which is consistent with their sphere-like shape, that for finite salt concentrations do not follow a clear sigmoidal trend. The lack of a sigmoidal trend is also consistent with their condensate's substantial deviations from a spherical shape (Figure 2). For no salt conditions, we fit the data to a sigmoidal function as given in eq 1.

$$C(r) = A_0 - A_1 \operatorname{erf}\left(\frac{r - R}{\sqrt{2} W}\right) \quad (1)$$

Here, r is the radial distance from the droplet center; erf is the error function; and A_0 , A_1 , R , and W are fit parameters. As noted in the work by Hummer and co-workers,²⁴ the density plateaus at the center for sufficiently large droplets with a defined density, where $A_0 + A_1$ and $A_0 - A_1$ will be the limiting concentrations, dense and dilute phase concentrations, respectively. For no salt

conditions, we were able to estimate the dilute and condensed phase concentrations from these limiting values. This approach works reasonably well for droplets that have spherical/ellipsoidal shapes. However, since the condensates for the conditions with finite salt concentrations significantly deviate from a sphere (they do not show a clear plateau as the center is approached), we used a surface reconstruction method⁵⁴ to estimate the volume and concentration instead of fitting the radial density profiles/using the limiting values. In the surface reconstruction approach, a virtual probe is used to define the accessible spatial volume, the inaccessible volume, and the surface between the two. We used OVITO software for this analysis.⁵⁵ We chose a 10 Å radius for the probe, which takes the nearest bead separation within the condensed phase into account. We also have tested a range of radii between 8 and 16 Å for no salt conditions and found that the difference between the condensed phase concentration estimations from two different methods is smallest for 10 Å probe for the small λ (Figure S3). In Table 1, condensed phase concentrations (obtained from two different techniques for no salt) are tabulated.

We report the protein concentration in the dense phase only until $\lambda = 1.03$ since we do not observe a viable cluster formation beyond $\lambda = 1.03$. As λ increases, the volume of the dense phase increases (and condensed phase concentration decreases accordingly) until the system is not capable of forming a

dense phase ($\lambda > 1.03$). The smallest protein concentration that we obtain in the condensed phase is 999.8 mg/mL (using surface reconstruction; or 720.8 using sigmoidal fit). We note that regardless of the analysis technique we used, even the smallest protein concentration that we obtain is still larger than the experimentally found protein concentration in the condensed phase. We discuss this further in the following subsection. We also report the concentrations in the dilute phase in Table 2.

Table 2. Protein Concentrations in the Dilute Phase Given in mg/mL (Surface Reconstruction)

	$\lambda = 1.000$	$\lambda = 1.010$	$\lambda = 1.020$	$\lambda = 1.025$	$\lambda = 1.030$
no salt	0	0	1.4	10.0	—
50 mM salt	0	0	1.7	11.3	30.8
100 mM salt	0	0	3.6	1.3	17.3
200 mM salt	0	0	0	2.7	10.7

As can be seen in Table 2, there is no protein in the dilute phase for $\lambda = 1$ and $\lambda = 1.01$ (for any salt condition), i.e., all proteins are in the dense phase. The coexistence of dense and dilute phases applies for $\lambda = 1.02$ and 1.025 for no salt conditions. Beyond $\lambda = 1.03$ there is no dense phase formation (for no salt conditions, there is no dense phase formation for $\lambda = 1.03$). The relative concentrations in the dense and dilute phases are discussed further in the following subsection.

We also found that water concentration in the dense phase increases as λ increases (Figure 3B). For the ionic species, we found that while they are concentrated in the dense phase (Figure 3C), there still is some ionic concentration in the dilute phase (including $\lambda = 1$). In summary, protein–protein interactions dominate the protein–water interactions at smaller

λ values, yielding very high protein concentrations in the dense phase and very small to no protein concentrations in the dilute phase. Since our goal is to optimize the λ until we establish a balance between protein–protein and protein–water interactions, we next calculated the excess transfer free energy based on the relative concentrations and compared our findings against experimental data.

Excess Transfer Free Energy and Relative Concentrations of the Condensed and Dilute Phases of FUS LC Condensates. Phase diagrams are quantitative descriptors of LLPS by defining the thermodynamic conditions at which distinct phases form (e.g., condensed and dilute phases) and coexist at equilibrium. Phase diagrams of LLPS are often defined on the temperature–concentration plane. Accordingly, coexistence data based on relative concentrations of condensed and dilute phases at a given temperature and pressure are metrics of particular interest. Benayad et al.²⁴ used a metric that takes this relative concentration into account to benchmark the scaled protein–protein interactions in the MARTINI 2 force field. They have used excess transfer energy, that is, the free energy associated with the transfer of protein chains from the dilute phase to the condensed droplet phase, formulated as

$$\Delta G = kT \ln \left(\frac{C_{\text{dilute}}}{C_{\text{dense}}} \right) \quad (2)$$

where k is the Boltzmann constant, T is the temperature, and C_{dilute} and C_{dense} are the concentrations in the dilute and dense phases, respectively. In their work, Benayad et al.,²⁴ for atmospheric pressure and room temperature, have identified a range of protein–protein scaling parameters that yield coexistence concentrations that agree well with the experimental data. In this work, we reported a similar experimental comparison for the scaled protein–water interactions. In Figure

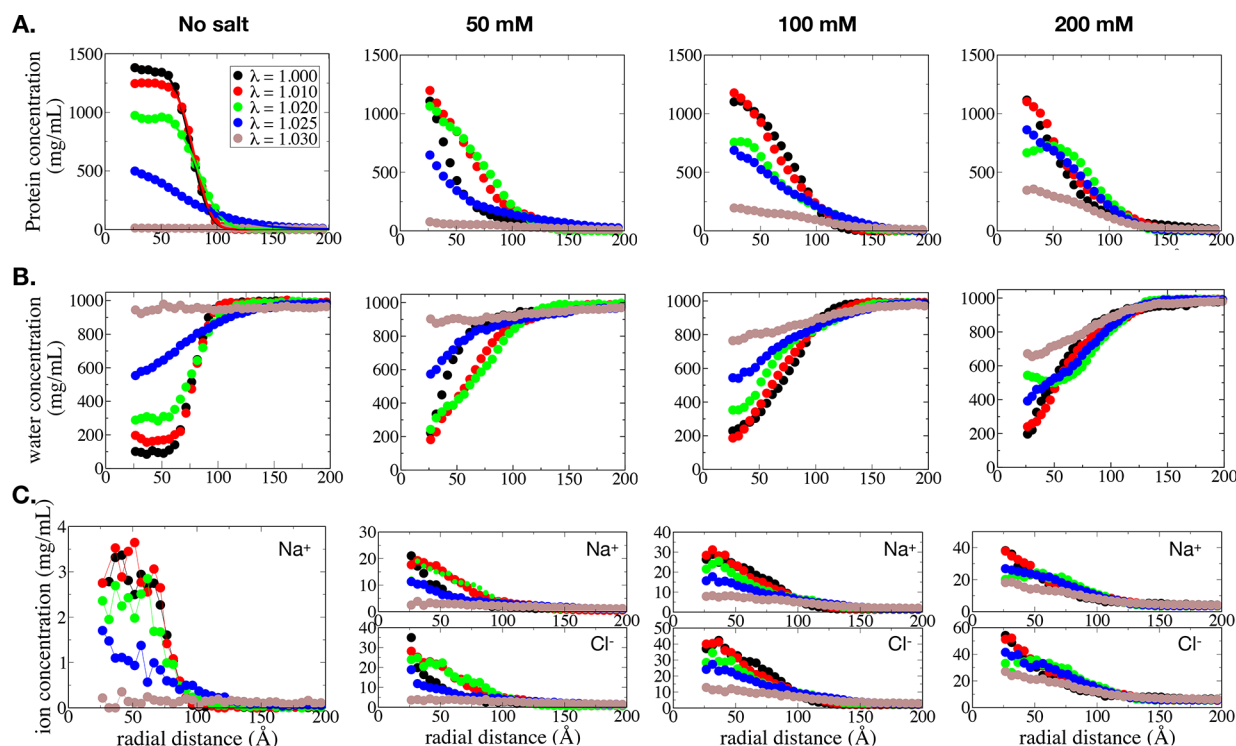


Figure 3. Radial concentration profiles of protein (A), water (B), and ions (C). From left to right, the first, second, third, and fourth columns are no salt, 50 mM salt, 100 mM salt, and 200 mM salt concentrations, respectively.

4A, we presented the protein concentration in the dense phase for different salt concentrations and scaling parameters (λ) that

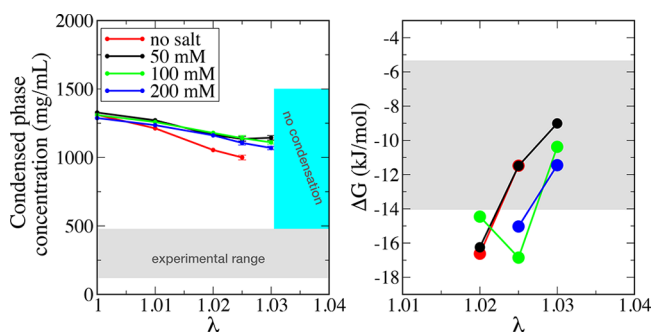


Figure 4. Relative protein concentrations in the dense phase and in the dilute phase as a function of the scaling parameters (λ): Protein concentration in the dilute phase (left) and transfer free energy (right). Gray-shaded areas are experimentally found ranges. See the text for experimental reference. Statistical errors are calculated by block averaging the data using two blocks. Error bars are smaller than the symbol size for most data points.

we tested. As noted in the previous section, dense phase volumes are estimated from a surface reconstruction technique. The general trend is a decrease in the dense phase protein concentration as λ increases, except for $\lambda = 1.03$ (smallest λ that forms a dense phase). Although our dense phase concentrations are larger than the experimentally identified dense phase concentrations (Figure 4A), we still found a range of protein–water scaling parameters for which the excess

transfer free energy agrees well with the experimental data (Figure 4B) within the range that we observe phase coexistence ($1.01 < \lambda \leq 1.03$). The experimental data range is referenced from the work by Fawzi and co-workers;^{36,39} where ref 36 measures the FUS LC concentration in the dense phase as approximately 120 mg/mL (spectroscopically) and in ref 39, a 477 mg/mL FUS LC concentration is deduced from the relative intensity of the glutamine side chain residue NMR resonances in the condensed phase (compared to a standard protein concentration in the dispersed phase, which is given as 100 μ M, or 1.71 mg/mL). 477 mg/mL FUS LC dense phase has been obtained from 15 mL samples of 1 mM FUS LC solutions³⁶ (from which we calculated the dilute phase concentration as approximately 14.3 mg/mL). We used these dense phase and their respective dilute phase concentrations to calculate the experimental range of transfer free energy (gray-shaded areas in Figure 4).

Contact Maps. We also calculated the interchain residual contact maps (square, symmetric matrices) at each λ parameter and salt concentration to quantitatively visualize individual contacts between different amino acids and contact clusters that hold the condensates together. Figure 5 reports the ensemble-averaged fraction of contacts between each residue i, j . We will analyze the data in Figure 5 in two dimensions: (i) as a function of the λ parameter, and (ii) as a function of salt concentration. Figure 5 shows that the contact propensity decreases as the λ parameter increases (from left to right), that is, as the protein–water interaction strengthens, at all salt concentrations. This observation is consistent with our findings reported in previous sections that the cluster formation weakens as the λ parameter

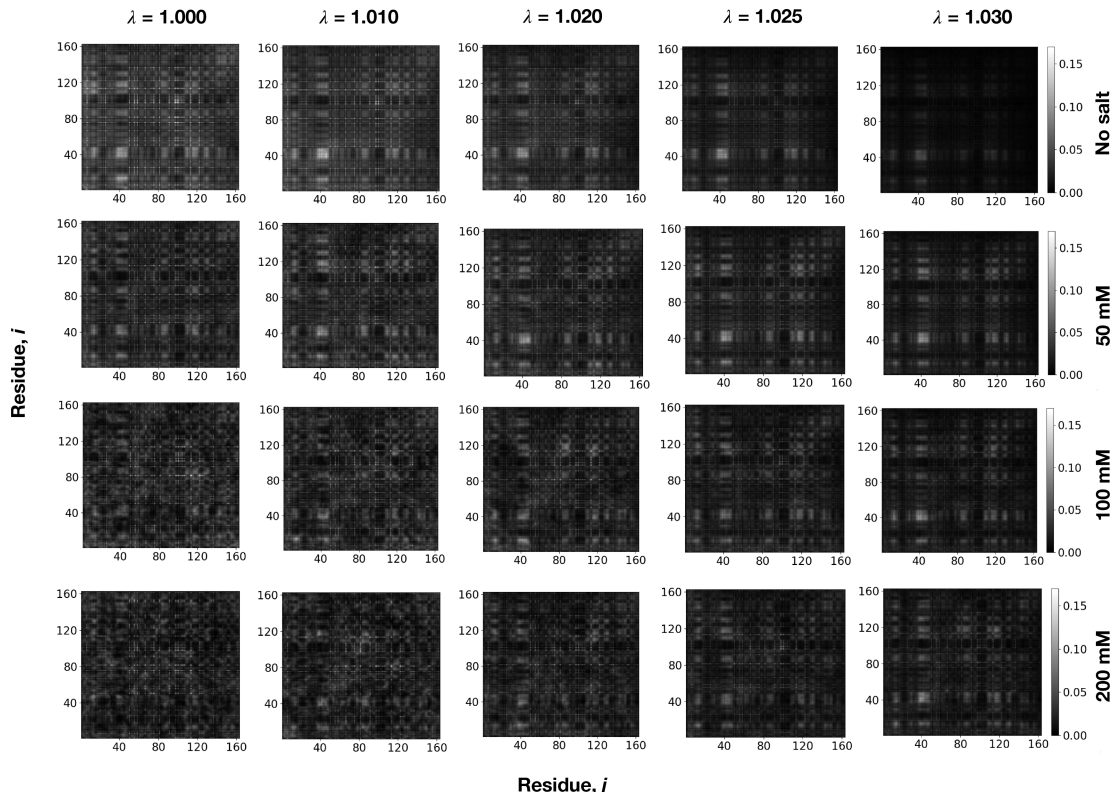


Figure 5. Fraction of the interchain residual contacts between each residue i, j . We first calculated a 16300×16300 binary matrix of contacts at each instantaneous frame (using a 0.8 nm cutoff), considering all protein chains in the system (we have 100 identical chains, each one is 163 residues). For each frame, we first averaged the contacts over 100 chains, reducing the matrix size to 163×163 , and then finally reported the ensemble average as the fraction of contact formation (the color scale from black to white).

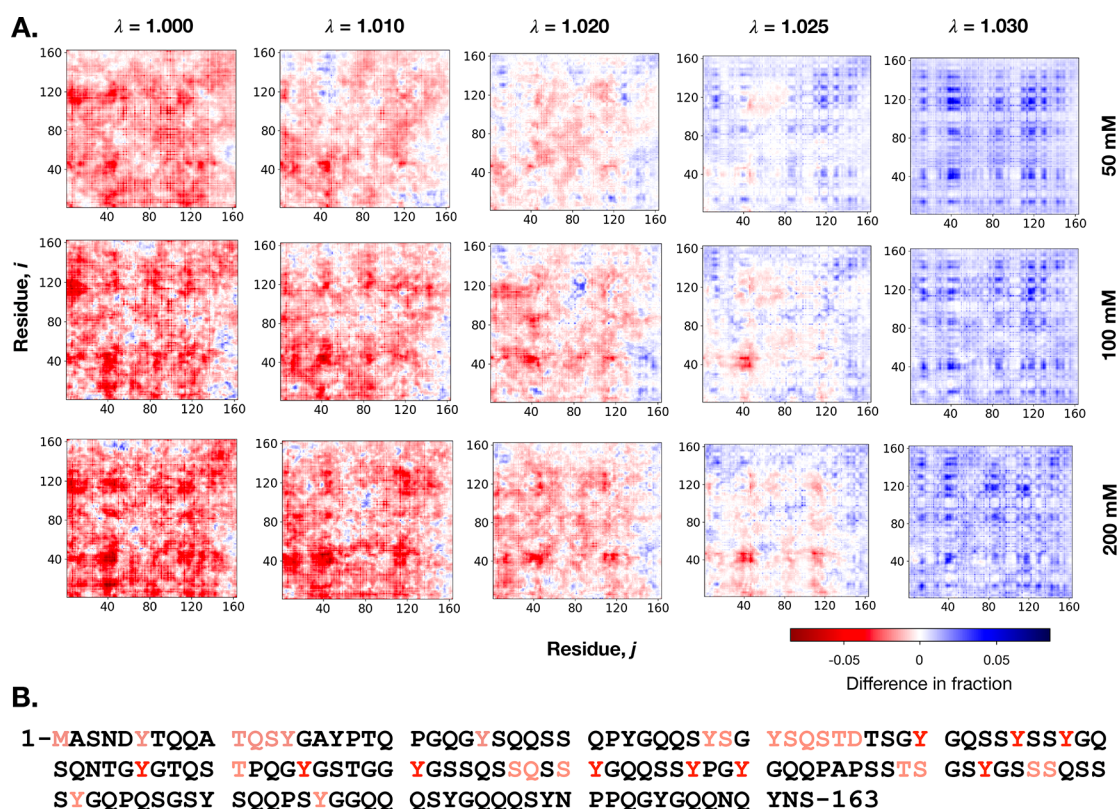


Figure 6. Summary of the effect of salt concentration on the interchain contacts. A. The difference in the fraction of the interchain residual contacts at various salt concentrations with respect to 0 mM salt. The color axis corresponds to (contact fraction_{X mM} − contact fraction_{0 mM}) where X is 50, 100, or 200 mM salt concentrations. B. FUS LC amino acid sequence. Amino acids with a high propensity to form contacts at 0 mM salt concentration for $\lambda = 1$ (fraction >0.1 in the top left corner of Figure 5) are shown in red font. The ones shown in faded red font are the ones that have fraction >0.1 at 0 mM salt concentration but lose the strength to have fraction <0.1 as the salt concentration increases (when $\lambda < 1.03$, as the salt makes all contacts stronger at $\lambda = 1.03$ for all salt concentrations).

increases at all salt concentrations. As the protein–protein contact formation weakens, the condensate formation also weakens. The majority of the strong contacts (fraction >0.1) are the tyrosine residues in (S/G)Y(S/G) repeating units of the FUS LC domain (Figure 6B). The 163 residue-long LC domain has 18 of these repeating units (SI Text). It has been previously shown that a progressive mutation of the central tyrosine residues to serine in (S/G)Y(S/G) consensus motif causes a systematic drop in the partitioning of the FUS LC domain to the stress granules.³⁷ Our findings are consistent with this experimental observation. We found that the lower the λ is (i.e., the larger the fraction of proteins in the largest cluster), the stronger the interactions of the central tyrosine residues in the (S/G)Y(S/G) consensus motif. Accordingly, we conclude that these central tyrosine residues play a major role in driving the dense phase formation.

Figure 6A shows the effect of salt concentration on interchain contact formation (from top to bottom for each λ). We found that the contact formation profile substantially changed upon the addition of salt. At $\lambda = 1$, where we found the largest concentration for the dense phase (Figure 4), the increase in salt concentration systematically reduces the contact fraction (Figure 6A, leftmost column). This reduction is more prominent in the fraction of cluster-type contacts (for example, between residues 35 to 50). Interestingly, however, contact clusters reappear in the presence of salt as the λ increases (Figure 5). FUS LC domain has characteristics of prion-like proteins, including being predominantly composed of uncharged amino

acids.⁵⁶ FUS LC domain has only 2 charged residues out of 163. These charged residues are located at positions 5 and 46, where we observe most of the reduction in the fraction of cluster-type contacts.

As the λ increases, however, the effect of salt reverses its trend, that is, the addition of salt makes the contact fraction weaker until $\lambda = 1.02$ – 1.025 , the change in contact fraction becomes more balanced at $\lambda = 1.02$ – 1.025 , but at $\lambda = 1.03$, all contacts become stronger with the addition of salt. Previously, a decreased solubility of FUS LC upon an increase in salt concentration was reported by Fawzi and co-workers.³⁶ The authors have reasoned that the salting-out of the hydrophobic residues (predominantly tyrosines) could be responsible for the decreased solubility. Our finding of the increase in protein–protein contacts at $\lambda = 1.03$ is consistent with this salting-out observation. Although we do not directly measure solubility, the salting-out effect means an effective increase in protein–protein interactions, which is consistent with the increase in amino acid contact fractions as we observe here. We note that the $\lambda = 1.03$ is the strength of protein–water interactions that gives an agreement in the transfer free energy with the experimental values for the systems with salt. Therefore, it is reasonable that we are observing the experimentally predicted effects at this specific λ , $\lambda = 1.03$. This observation provides another layer of validation to our modeling approach.

For the smaller λ values, protein–protein interactions dominate the protein–water interactions. As opposed to the net increase in amino acid contacts (i.e., salting-out) observed at

a higher λ ($\lambda = 1.03$), we observed an effective reduction in amino acid contact fraction at a lower λ (Figure 6), which is rather consistent with electrostatic screening. We note that even though protein–protein contact fractions become lower upon the addition of salt (compared to 0 mM salt) at a low λ , this reduction neither causes dissociation of condensates (Figure 1, blue data, left column) nor a significant change in the protein concentration in the dense phase (Table 1). There are, however, significant changes to the condensate morphology. An increased salt concentration at a low λ yields percolated nonspherical protein condensates instead of spherical droplets (Figure 2). The shape of the condensed phase is governed by surface tension. Surface tension (or interfacial tension) arises from the balance between intermolecular forces.⁵⁷ If the cohesive forces are stronger than the adhesive forces, protein molecules will collapse (or coalesce) into a condensed phase due to the inward component of the net force. Moreover, this condensed phase will also tend to shrink its interface due to the tangential component of the net force. Interfacial tension causes the biomolecular condensates to minimize their surface area with the dilute phase, driving them toward a spherical shape (since a sphere provides the minimum surface for a given volume). Accordingly, we argue that the deviations from spherical shapes at elevated salt concentrations (at low λ values, especially $\lambda = 1$) may be explained by lower surface tension that arises from effective protein–protein interactions being reduced with the presence of salt.

Finally, we also examined the contact formation for the case of 50 mM added salt that starts from a preformed droplet (see the Identification of Condensate Formation subsection for the description). As presented in Figure S5, we found that the initially spherical droplet remains largely spherical within the simulation time (never forms rod-like percolated structures) for this case. Therefore, this case helps us assess the effect of salt concentration on the interchain contacts without potentially confounding factors from different dense phase morphologies. Figure S7 shows both the contact propensity (A) and the effect of salt concentration (B) on the contact propensity. Figure S7A shows that the contact propensity decreases as the λ parameter increases, similar to the findings in Figure 5. Figure S7B shows, however, that the change in contact fraction with respect to 0 mM salt at $\lambda = 1$ is weaker (resembling $\lambda = 1.02$ at 50 mM salt in Figure 6A), although the salting out effect at high λ values ($\lambda = 1.025$ and 1.03) are more prominent and stronger compared to those in Figure 6A.

CONCLUSIONS

In this work, we studied the condensate formation of the FUS LC domain at various salt concentrations using a MARTINI 3.0 force field by systematically scaling the interaction strength of protein–water beads, introducing a parameter, λ . The λ is a factor that we multiply the ϵ of the Lennard-Jones interaction between protein–water beads, that is, the larger the λ is, the stronger the protein–water interactions are, and $\lambda = 1$ is the base strength of protein–water interactions. We systematically increased the λ in order to reproduce the condensate properties, specifically the excess free energy of transferring protein chains from the dilute phase to the condensed phase. While we do not observe a dilute phase at weak protein–water interactions (low λ), that is, all protein chains coalesce into one dense phase when the protein–water interactions are unmodified, or when the $\lambda < 1.02$. As we scaled λ up, we started observing a phase equilibrium. We presented the results from each scaling at

various salt concentrations. The optimal λ is slightly different in the presence or in the absence of the salt. However, within the concentration range of the salt that we tested for the FUS LC domain, the optimal λ does not change at different salt concentrations; it only changes based on whether we have salt in the system or not. We note that the optimal value of the scaling λ parameter applies specifically to FUS LC. The optimal scaling of the λ parameter might depend on the sequence characteristics of a given protein. This work describes an *ad hoc* optimization protocol in an attempt to rebalance protein–water interactions in this force field to reproduce the experimental data of FUS LC condensation. Here, we systematically tested this parameter for one specific protein for the widespread use of the force field in protein phase separation studies. While the optimal parameters obtained for this protein may not be transferable to other proteins, they should still serve as a good initial crude estimate for other phase separation disordered systems.

We also found that the presence of salt concentration affects the dense phase morphology, especially when protein–water interactions are weaker (low λ). While a more extensive exploration of the effect of salt concentration on condensate morphology is a part of future work, here, we showed that the individual amino acid contacts are highly sensitive to salt concentration, which might be related to the nonspherical morphology. Salt has more predominant screening effects at low λ values, whereas the opposite salting-out effects become more predominant as λ increases. While the screening effects do not cause dissociation of the condensates, we argued that the screening may be responsible for condensates losing their spherical shape by reducing the surface tension. At high λ values, we did not observe any obvious effects of salt on morphology, where the salting-out effects were more predominant. In summary, our findings show that there is a balance between electrostatic screening effects and salting-out effects and this balance is highly sensitive to the strength of protein–water interactions (λ). We also note that we observed an initial condition dependence of the morphology at low λ conditions at 50 mM salt. This finding emphasizes the necessity of future work for exploring condensate morphology with proper advanced sampling techniques.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.2c01273>.

FUS LC sequence, snapshot of an example monodispersed initial configuration, total potential energy of systems as a function of time and its running average, fraction of proteins in three most populating clusters for the $\lambda = 1.04$ at no salt conditions starting from a preformed droplet, time dependent change in the radius of gyration and the ratio of smallest and largest eigenvalues of the gyration tensor of the droplet, analysis of the effects of 50 mM added salt on the initially preformed droplet, protein concentration in the condensed phase analyzed by two different techniques, summary of the contact formation in 50 mM salt simulations starting from a preformed droplet, and equilibration time periods for all systems (PDF)

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Notes

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REFERENCES

- (1) Brangwynne, C. P.; Eckmann, C. R.; Courson, D. S.; Rybarska, A.; Hoeghe, C.; Gharakhani, J.; Jülicher, F.; Hyman, A. A. Germline P granules are liquid droplets that localize by controlled dissolution/condensation. *Science* **2009**, *324*, 1729–1732.
- (2) Banani, S. F.; Lee, H. O.; Hyman, A. A.; Rosen, M. K. Biomolecular condensates: organizers of cellular biochemistry. *Nat. Rev. Mol. Cell Biol.* **2017**, *18*, 285–298.
- (3) Lyon, A. S.; Peeples, W. B.; Rosen, M. K. A framework for understanding the functions of biomolecular condensates across scales. *Nat. Rev. Mol. Cell Biol.* **2021**, *22*, 215–235.
- (4) Wang, B.; Zhang, L.; Dai, T.; Qin, Z.; Lu, H.; Zhang, L.; Zhou, F. Liquid–liquid phase separation in human health and diseases. *Signal Transduction Targeted Ther.* **2021**, *6*, 1–16.
- (5) Brangwynne, C. P.; Mitchison, T. J.; Hyman, A. A. Active liquid-like behavior of nucleoli determines their size and shape in *Xenopus laevis* oocytes. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108*, 4334–4339.
- (6) Li, P.; Banjade, S.; Cheng, H.-C.; Kim, S.; Chen, B.; Guo, L.; Llaguno, M.; Hollingsworth, J. V.; King, D. S.; Banani, S. F.; et al. Phase transitions in the assembly of multivalent signalling proteins. *Nature* **2012**, *483*, 336–340.
- (7) Nott, T. J.; Petsalaki, E.; Farber, P.; Jervis, D.; Fussner, E.; Plochowietz, A.; Craggs, T. D.; Bazett-Jones, D. P.; Pawson, T.; Forman-Kay, J. D.; et al. Phase transition of a disordered nuage protein generates environmentally responsive membraneless organelles. *Molecular cell* **2015**, *57*, 936–947.
- (8) Lin, Y.; Protter, D. S.; Rosen, M. K.; Parker, R. Formation and maturation of phase-separated liquid droplets by RNA-binding proteins. *Molecular cell* **2015**, *60*, 208–219.
- (9) Molliex, A.; Temirov, J.; Lee, J.; Coughlin, M.; Kanagaraj, A. P.; Kim, H. J.; Mittag, T.; Taylor, J. P. Phase separation by low complexity domains promotes stress granule assembly and drives pathological fibrillization. *Cell* **2015**, *163*, 123–133.
- (10) Feric, M.; Vaidya, N.; Harmon, T. S.; Mitrea, D. M.; Zhu, L.; Richardson, T. M.; Kriwacki, R. W.; Pappu, R. V.; Brangwynne, C. P. Coexisting liquid phases underlie nucleolar subcompartments. *Cell* **2016**, *165*, 1686–1697.
- (11) Strom, A. R.; Emelyanov, A. V.; Mir, M.; Fyodorov, D. V.; Darzacq, X.; Karpen, G. H. Phase separation drives heterochromatin domain formation. *Nature* **2017**, *547*, 241–245.
- (12) Sabari, B. R.; Dall'Agnese, A.; Boija, A.; Klein, I. A.; Coffey, E. L.; Shrinivas, K.; Abraham, B. J.; Hannett, N. M.; Zamudio, A. V.; Manteiga, J. C.; Li, C. H.; Guo, Y. E.; Day, D. S.; Schuijers, J.; Vasile, E.; Malik, S.; Hnisz, D.; Lee, T. I.; Cisse, I.; Roeder, R. G.; Sharp, P. A.; Chakraborty, A. K.; Young, R. A. Coactivator condensation at super-enhancers links phase separation and gene control. *Science* **2018**, *361*, eaar3958.
- (13) Boija, A.; Klein, I. A.; Sabari, B. R.; Dall'Agnese, A.; Coffey, E. L.; Zamudio, A. V.; Li, C. H.; Shrinivas, K.; Manteiga, J. C.; Hannett, N. M.; Abraham, B. J.; Afeyan, L. K.; Guo, Y. E.; Rimel, J. K.; Fant, C. B.; Schuijers, J.; Lee, T. I.; Taatjes, D. J.; Young, R. A. Transcription factors activate genes through the phase-separation capacity of their activation domains. *Cell* **2018**, *175*, 1842–1855.
- (14) Dignon, G. L.; Zheng, W.; Mittal, J. Simulation methods for liquid–liquid phase separation of disordered proteins. *Current opinion in chemical engineering* **2019**, *23*, 92–98.
- (15) Dignon, G. L.; Zheng, W.; Kim, Y. C.; Best, R. B.; Mittal, J. Sequence determinants of protein phase behavior from a coarse-grained model. *PLoS computational biology* **2018**, *14*, e1005941.
- (16) Ruff, K. M.; Harmon, T. S.; Pappu, R. V. CAMELOT: A machine learning approach for coarse-grained simulations of aggregation of block-copolymeric protein sequences. *J. Chem. Phys.* **2015**, *143*, 243123.
- (17) Statt, A.; Casademunt, H.; Brangwynne, C. P.; Panagiotopoulos, A. Z. Model for disordered proteins with strongly sequence-dependent liquid phase behavior. *J. Chem. Phys.* **2020**, *152*, 075101.
- (18) Das, S.; Lin, Y.-H.; Vernon, R. M.; Forman-Kay, J. D.; Chan, H. S. Comparative roles of charge, π , and hydrophobic interactions in sequence-dependent phase separation of intrinsically disordered proteins. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117*, 28795–28805.
- (19) Latham, A. P.; Zhang, B. Consistent force field captures homologue-resolved HP1 phase separation. *J. Chem. Theory Comput.* **2021**, *17*, 3134–3144.
- (20) Dannenhoffer-Lafage, T.; Best, R. B. A data-driven hydrophobicity scale for predicting liquid–liquid phase separation of proteins. *J. Phys. Chem. B* **2021**, *125*, 4046–4056.
- (21) Garaizar, A.; Espinosa, J. R. Salt dependent phase behavior of intrinsically disordered proteins from a coarse-grained model with explicit water and ions. *J. Chem. Phys.* **2021**, *155*, 125103.
- (22) Tsanai, M.; Frederix, P. W.; Schroer, C. F.; Souza, P. C.; Marrink, S. J. Coacervate formation studied by explicit solvent coarse-grain molecular dynamics with the Martini model. *Chem. Sci.* **2021**, *12*, 8521–8530.
- (23) Tesei, G.; Schulze, T. K.; Crehuet, R.; Lindorff-Larsen, K. Accurate model of liquid–liquid phase behavior of intrinsically disordered proteins from optimization of single-chain properties. *Proc. Natl. Acad. Sci. U. S. A.* **2021**, *118*. DOI: 10.1073/pnas.2111696118
- (24) Benayad, Z.; von Bulow, S.; Stelzl, L. S.; Hummer, G. Simulation of FUS protein condensates with an adapted coarse-grained model. *J. Chem. Theory Comput.* **2021**, *17*, 525–537.
- (25) Joseph, J. A.; Reinhardt, A.; Aguirre, A.; Chew, P. Y.; Russell, K. O.; Espinosa, J. R.; Garaizar, A.; Collepardo-Guevara, R. Physics-driven coarse-grained model for biomolecular phase separation with near-quantitative accuracy. *Nature Computational Science* **2021**, *1*, 732–743.
- (26) Marrink, S. J.; Risselada, H. J.; Yefimov, S.; Tieleman, D. P.; De Vries, A. H. The MARTINI force field: coarse grained model for biomolecular simulations. *J. Phys. Chem. B* **2007**, *111*, 7812–7824.
- (27) Monticelli, L.; Kandasamy, S. K.; Periole, X.; Larson, R. G.; Tieleman, D. P.; Marrink, S.-J. The MARTINI coarse-grained force field: extension to proteins. *J. Chem. Theory Comput.* **2008**, *4*, 819–834.
- (28) Machado, N.; Bruininks, B. M.; Singh, P.; Dos Santos, L.; Dal Pizzol, C.; Dieamant, G. d. C.; Kruger, O.; Martin, A. A.; Marrink, S. J.; Souza, P. C.; et al. Complex nanoemulsion for vitamin delivery: droplet organization and interaction with skin membranes. *Nanoscale* **2022**, *14*, 506–514.
- (29) Grünewald, F.; Kroon, P. C.; Souza, P. C.; Marrink, S. J. *Structural Genomics*; Springer, 2021; pp 315–335.

- (30) Roel-Touris, J.; Jiménez-García, B.; Bonvin, A. M. Integrative modeling of membrane-associated protein assemblies. *Nat. Commun.* **2020**, *11*, 1–11.
- (31) Nerenberg, P. S.; Jo, B.; So, C.; Tripathy, A.; Head-Gordon, T. Optimizing solute–water van der Waals interactions to reproduce solvation free energies. *J. Phys. Chem. B* **2012**, *116*, 4524–4534.
- (32) Best, R. B.; Zheng, W.; Mittal, J. Balanced protein–water interactions improve properties of disordered proteins and non-specific protein association. *J. Chem. Theory Comput.* **2014**, *10*, 5113–5124.
- (33) Larsen, A. H.; Wang, Y.; Bottaro, S.; Grudinin, S.; Arleth, L.; Lindorff-Larsen, K. Combining molecular dynamics simulations with small-angle X-ray and neutron scattering data to study multi-domain proteins in solution. *PLoS computational biology* **2020**, *16*, e1007870.
- (34) Thomasen, F. E.; Pesce, F.; Roesgaard, M. A.; Tesi, G.; Lindorff-Larsen, K. Improving the global dimensions of intrinsically disordered proteins in Martini 3. *J. Chem. Theory Comput.* **2022**, *18*, 2033.
- (35) Thomasen, F. E.; Pesce, F.; Roesgaard, M. A.; Tesi, G.; Lindorff-Larsen, K. Improving Martini 3 for Disordered and Multidomain Proteins. *J. Chem. Theory Comput.* **2022**, *18*, 2033–2041.
- (36) Burke, K. A.; Janke, A. M.; Rhine, C. L.; Fawzi, N. L. Residue-by-residue view of in vitro FUS granules that bind the C-terminal domain of RNA polymerase II. *Molecular cell* **2015**, *60*, 231–241.
- (37) Kato, M.; Han, T. W.; Xie, S.; Shi, K.; Du, X.; Wu, L. C.; Mirzaei, H.; Goldsmith, E. J.; Longgood, J.; Pei, J.; et al. Cell-free formation of RNA granules: low complexity sequence domains form dynamic fibers within hydrogels. *Cell* **2012**, *149*, 753–767.
- (38) Wang, J.; Choi, J.-M.; Holehouse, A. S.; Lee, H. O.; Zhang, X.; Jahnel, M.; Maharana, S.; Lemaitre, R.; Pozniakovsky, A.; Drechsel, D.; Poser, I.; Pappu, R.; Alberti, S.; Hyman, A. A molecular grammar governing the driving forces for phase separation of prion-like RNA binding proteins. *Cell* **2018**, *174*, 688–699.
- (39) Murthy, A. C.; Dignon, G. L.; Kan, Y.; Zerze, G. H.; Parekh, S. H.; Mittal, J.; Fawzi, N. L. Molecular interactions underlying liquid–liquid phase separation of the FUS low-complexity domain. *Nature structural & molecular biology* **2019**, *26*, 637–648.
- (40) Souza, P. C.; Alessandri, R.; Barnoud, J.; Thallmair, S.; Faustino, I.; Grünewald, F.; Patmanidis, I.; Abdizadeh, H.; Bruininks, B. M.; Wassenaar, T. A.; Kroon, P. C.; Melcr, J.; Nieto, V.; Corradi, V.; Khan, H. M.; Domanski, J.; Javanainen, M.; Martinez-Seara, H.; Reuter, N.; B. B. R.; Vattulainen, L.; Monticelli, L.; Periole, X.; Tieleman, D. P.; de Vries, A. H.; Marrink, S. J. Martini 3: a general purpose force field for coarse-grained molecular dynamics. *Nat. Methods* **2021**, *18*, 382–388.
- (41) Retrieved from: Martini 3 Tutorials. <http://cgmartini.nl/index.php/martini-3-tutorials>, <http://cgmartini.nl/index.php/martini3beta>, accessed: 2021-06-30.
- (42) Robustelli, P.; Piana, S.; Shaw, D. E. Developing a molecular dynamics force field for both folded and disordered protein states. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115*, E4758–E4766.
- (43) Bussi, G.; Donadio, D.; Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **2007**, *126*, 014101.
- (44) Parrinello, M.; Rahman, A. Polymorphic transitions in single crystals: A new molecular dynamics method. *J. Appl. Phys.* **1981**, *52*, 7182–7190.
- (45) Tironi, I. G.; Sperb, R.; Smith, P. E.; van Gunsteren, W. F. A generalized reaction field method for molecular dynamics simulations. *J. Chem. Phys.* **1995**, *102*, 5451–5459.
- (46) Liu, Y.; Panagiotopoulos, A. Z.; Debenedetti, P. G. Low-temperature fluid-phase behavior of ST2 water. *J. Chem. Phys.* **2009**, *131*, 104508.
- (47) Berendsen, H. J.; van der Spoel, D.; van Drunen, R. GROMACS: A message-passing parallel molecular dynamics implementation. *Comput. Phys. Commun.* **1995**, *91*, 43–56.
- (48) Hess, B.; Kutzner, C.; Van Der Spoel, D.; Lindahl, E. GROMACS 4: Algorithms for highly efficient, load-balanced, and scalable molecular simulation. *J. Chem. Theory Comput.* **2008**, *4*, 435–447.
- (49) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, *1*, 19–25.
- (50) Van Rossum, G.; Drake, F. L. *Python 3 Reference Manual*; CreateSpace: Scotts Valley, CA, 2009.
- (51) Chatterjee, S.; Kan, Y.; Brzezinski, M.; Koynov, K.; Regy, R. M.; Murthy, A. C.; Burke, K. A.; Michels, J. J.; Mittal, J.; Fawzi, N. L.; Parekh, S. Reversible kinetic trapping of FUS biomolecular condensates. *Advanced Science* **2022**, *9*, 2104247.
- (52) Garaizar, A.; Espinosa, J. R.; Joseph, J. A.; Collepardo-Guevara, R. Kinetic interplay between droplet maturation and coalescence modulates shape of aged protein condensates. *Sci. Rep.* **2022**, *12*, 1–13.
- (53) Linsenmeier, M.; Hondele, M.; Grigolato, F.; Secchi, E.; Weis, K.; Arosio, P. Dynamic arrest and aging of biomolecular condensates are modulated by low-complexity domains, RNA and biochemical activity. *Nat. Commun.* **2022**, *13*, 3030.
- (54) Stukowski, A. Computational analysis methods in atomistic modeling of crystals. *Jom* **2014**, *66*, 399–407.
- (55) Stukowski, A. Visualization and analysis of atomistic simulation data with OVITO—the Open Visualization Tool. *Modell. Simul. Mater. Sci. Eng.* **2010**, *18*, 015012.
- (56) King, O. D.; Gitler, A. D.; Shorter, J. The tip of the iceberg: RNA-binding proteins with prion-like domains in neurodegenerative disease. *Brain research* **2012**, *1462*, 61–80.
- (57) Israelachvili, J. N. *Intermolecular and surface forces*; Academic Press, 2011.